

Adversarial AI in Games: Monte Carlo Simulation for Briscas

Abstract

This project studies the use of Monte Carlo simulation in the card game *Briscas*. Since the game involves hidden information and random card distributions, it provides a good environment for adversarial AI techniques based on uncertainty. The implemented agent evaluates possible actions by simulating many possible continuations of the game and estimating which move leads to better outcomes.

Several experiments were performed against random and heuristic agents. The results show that the Monte Carlo agent consistently performs better than baseline strategies, especially when paired with another Monte Carlo agent in team configurations.

1. Introduction

The objective of this project was to design and evaluate an adversarial AI agent capable of playing *Briscas*, a team-based Spanish card game with hidden information. Unlike deterministic games such as chess or checkers, *Briscas* contains uncertainty because players cannot observe the cards held by opponents. This makes the game a good example of a stochastic and partially observable environment.

The main problem addressed in this assignment is how an AI agent can make reasonable decisions even when the full game state is unknown. Traditional adversarial search methods like Minimax become difficult to apply because the branching factor grows very quickly and the agent does not know the exact state of the game.

For this reason, the project focuses on Monte Carlo simulation. Instead of trying to compute the perfect move through exhaustive search, the agent estimates which action is best by simulating many possible continuations of the game and averaging the outcomes.

Another important aspect of this assignment is the team component. Since *Briscas* is played in pairs, performance depends not only on the strength of the individual agent but also on how well it performs with different teammates.

2. Why Monte Carlo?

Several approaches were considered before deciding to use Monte Carlo simulation.

Approach	Main Limitation
Deep Neural Networks	Requires large training datasets and significant training time
Averaging over Clairvoyance	Assumes complete information after the deal

Minimax with Belief States	Computationally too expensive
Monte Carlo Simulation	Handles uncertainty naturally through sampling

Although Deep Neural Networks were initially considered, they introduced a level of complexity that was unnecessary for the scale of this assignment. Training a strong neural agent would require generating large datasets, tuning hyperparameters, and running many training iterations. In contrast, Monte Carlo simulation works reasonably well without a long training phase.

Minimax with belief states was also considered. However, Briscas contains hidden information and many possible future states, so the search tree becomes extremely large. Even shallow searches can become computationally expensive.

Monte Carlo simulation appeared to be the most practical choice because it works directly with uncertainty. Instead of assuming complete information, the agent generates possible hidden states that are consistent with the cards already observed during the game. The rest of the hand is then simulated many times to estimate the quality of each possible move.

Another advantage is that Monte Carlo methods are relatively flexible. Even with simple rollout strategies, the agent can still produce decent gameplay without needing handcrafted evaluation functions.

3. The Briscas Game

Briscas is a four-player trick-taking card game played in teams. Players 0 and 2 form one team, while players 1 and 3 form the opposing team.

The game uses a 40-card Spanish deck divided into four suits:

- Oros
- Copas
- Espadas
- Bastos

The cards are numbered from 1–7 and 10–12. Some cards are worth more points than others. For example:

- Ace = 11 points
- Three = 10 points
- King = 4 points
- Knight = 3 points

- Jack = 2 points

At the beginning of the game, each player receives three cards and a trump suit is revealed. Players take turns playing cards into tricks. If possible, players must follow the leading suit; otherwise, they may play any card.

The winner of the trick is determined by either:

- the highest trump card played, or
- the highest card of the leading suit if no trump appears.

One reason Briscas is interesting for AI is that the game is only partially observable. A player knows:

- their own hand,
- the cards already played,
- and the trump suit,

but the opponent hands remain hidden throughout the game.

This uncertainty makes perfect planning impossible, so the AI must estimate likely situations rather than compute exact future states.

4. Monte Carlo Agent Design

The implemented agent uses Monte Carlo planning based on hidden-state sampling.

The game state includes:

- the current hand of the agent,
- cards already played,
- the active trump suit,
- current team scores,
- and the history of previous tricks.

For every legal move available, the agent performs a sequence of simulations.

First, the algorithm generates random assignments of the unknown cards to the other players. These assignments must remain compatible with all visible information. For example, cards already played cannot appear again in sampled opponent hands.

After generating a possible hidden state, the rest of the game is simulated using a rollout policy. In this implementation, the rollout strategy is mostly random, although it still respects the legal rules of the game.

The outcome of each simulation is recorded. If the simulated team wins or achieves a better final score, the corresponding action receives a positive evaluation.

The estimated value of an action is computed as:

$$V(a) = \frac{1}{N} \sum_{i=1}^N R_i$$

where:

- (N) is the number of simulations,
- and (R_i) is the result of simulation (i).

Finally, the action with the highest average value is selected.

One important observation is that the agent does not “learn” in the same way as a reinforcement learning system. Instead, its strength comes from repeatedly estimating future outcomes through simulation. Increasing the number of simulations generally improves performance, although it also increases computation time.

5. Experimental Evaluation

The project requirements emphasized evaluation, not only implementation. For this reason, several experiments were performed against different types of opponents and teammates.

The main metric used was win rate over multiple games. Each configuration was tested over approximately 200 matches to reduce randomness in the results.

The following configurations were evaluated:

Match Configuration	Win Rate
(MC + MC) vs (Random + Random)	0.770
(MC + Heuristic) vs (Heuristic + Heuristic)	0.580
(MC + MC) vs (Heuristic + Heuristic)	0.665

The first result shows that the Monte Carlo agents clearly outperform random gameplay. This was expected, but it still demonstrates that the simulations provide useful strategic information even with relatively simple rollout policies.

One interesting observation from the experiments is that the Monte Carlo agents performed noticeably better when paired together than when paired with heuristic agents. The difference between:

- (MC + Heuristic) and

- (MC + MC)

suggests that partner quality has a significant effect on team performance.

This fulfills the “partner sensitivity” requirement from the assignment. Even without explicit communication between agents, two Monte Carlo agents appear to coordinate better simply because they evaluate situations using similar strategies.

Another observation is that the heuristic agents still performed reasonably well. Monte Carlo did not dominate every game completely, which is expected in stochastic card games where luck also plays a role.

Overall, the experiments suggest that Monte Carlo planning produces robust behavior across different conditions and opponent types.

6. Limitations and Future Work

The current implementation still has several limitations.

The largest limitation is the rollout policy. During simulations, most actions are selected randomly. This means that the simulated games are not always strategically realistic. A stronger rollout policy could likely improve performance significantly.

Another limitation is computational cost. Increasing the number of simulations improves the quality of decisions, but also increases execution time. There is a practical trade-off between speed and accuracy.

Future work could explore:

- heuristic-guided rollouts,
- Upper Confidence Bounds for Trees (UCT),
- reinforcement learning integration,
- or hybrid Monte Carlo + neural approaches.

It would also be interesting to test the agent against stronger human players or against more advanced AI opponents.

7. Conclusion

This project explored the use of Monte Carlo simulation for the card game Briscas, treating the game as a stochastic and partially observable adversarial environment. The results showed that even relatively simple Monte Carlo planning can outperform random and heuristic strategies without requiring expensive training procedures.

One of the most interesting aspects of the experiments was the effect of teammate quality on performance. The agent consistently achieved better results when paired

with another Monte Carlo agent, suggesting that implicit coordination emerges naturally from similar decision-making strategies.

Although the implementation still has limitations, particularly in the rollout policy, the project demonstrates that Monte Carlo methods are a practical approach for imperfect-information games and provide a solid balance between performance and implementation complexity.

References

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